Scoreboard

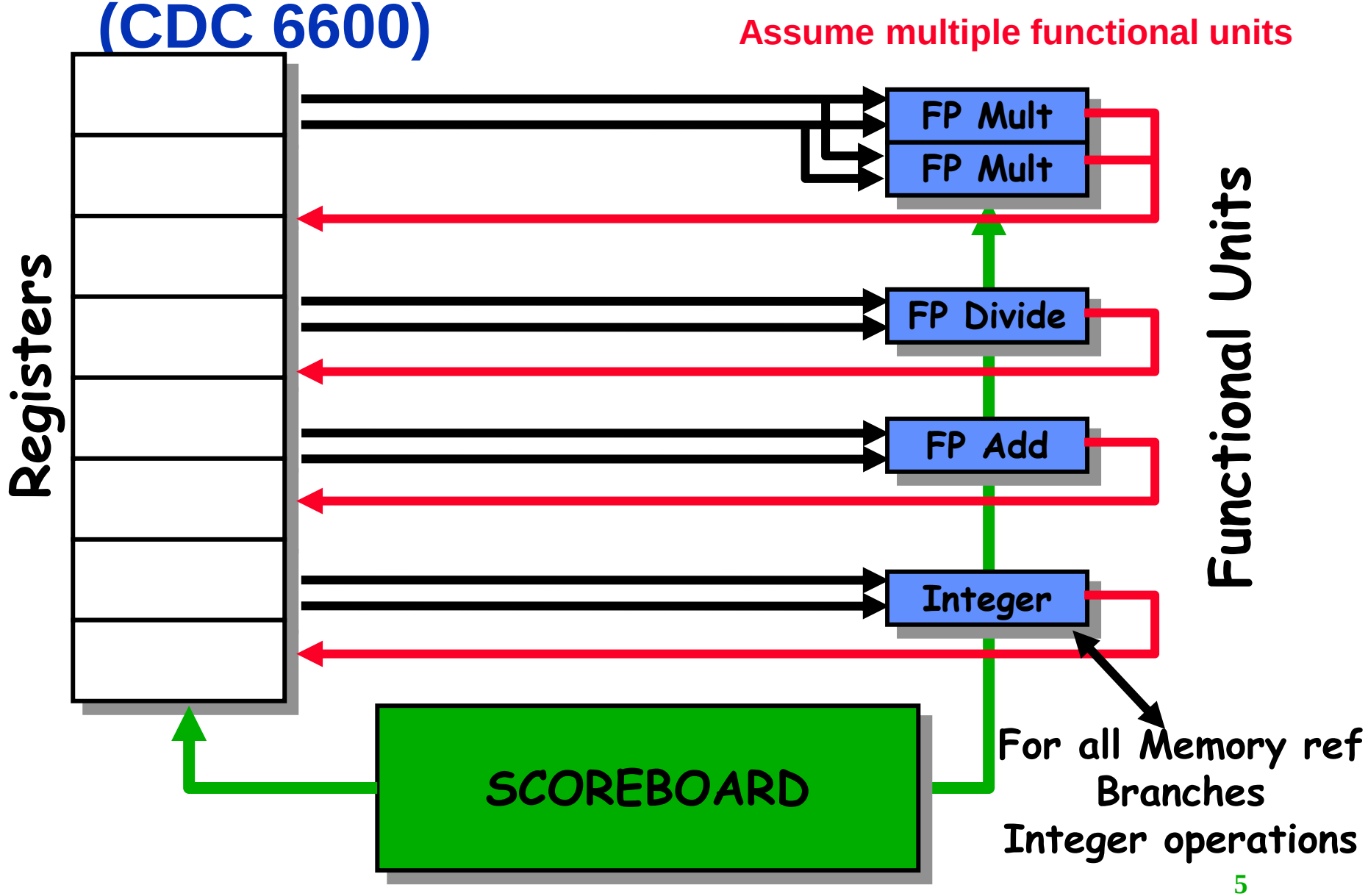
Scoreboard: a bookkeeping technique

- **Out-of-order execution divides ID stage:**
  1. **Issue—decode instructions, check for structural hazards**
  2. **Read operands—wait until no data hazards, then read operands**
- **Scoreboards date to CDC6600 in 1963**
- **Instructions execute whenever not dependent on previous instructions and no hazards.**
- **CDC 6600: In order issue, out-of-order execution, out-of-order commit (or completion)**
  - **No forwarding!**

Scoreboard Implications

- Out-of-order completion => WAR, WAW hazards
- Solutions for WAR:
  - Stall writeback until registers have been read
  - Read registers only during Read Operands stage
- Solution for WAW:
  - Detect hazard and stall issue of new instruction until other instruction completes
- No register renaming!
- Need to have multiple instructions in execution phase => multiple execution units or pipelined execution units
- Scoreboard keeps track of dependencies between instructions that have already issued.
- Scoreboard replaces ID, EX, WB with 4 stages

Scoreboard Architecture (CDC 6600)



Four Stages of Scoreboard Control

- **Issue—decode instructions & check for structural hazards (ID1)**
  - **Instructions issued in program order (for hazard checking)**
  - **Don't issue if structural hazard**
  - **Don't issue if instruction is output dependent on any previously issued but uncompleted instruction (no WAW hazards)**
- **Read operands—wait until no data hazards, then read operands (ID2)**
  - **All real dependencies (RAW hazards) resolved in this stage, since we wait for instructions to write back data.**
  - **No forwarding of data in this model**

Four Stages of Scoreboard Control

- **Execution—operate on operands (EX)**
  - The functional unit begins execution upon receiving operands. When the result is ready, it notifies the scoreboard that it has completed execution.
- **Write result—finish execution (WB)**
  - Stall until no WAR hazards with previous instructions:

Example:

```
DIVD   F0,F2,F4
ADDD   F10,F0,F8
SUBD   F8,F8,F14
```

CDC 6600 scoreboard would stall SUBD until ADDD reads operands

Three Parts of the Scoreboard

- **Instruction status:**
  Which of 4 steps the instruction is in

- **Functional unit status:—Indicates the state of the functional unit (FU). 9 fields for each functional unit**

Busy:	**Indicates whether the unit is busy or not**
Op:	**Operation to perform in the unit (e.g., + or –)**
Fi:	**Destination register**
Fj,Fk:	**Source-register numbers**
Qj,Qk:	**Functional units producing source registers Fj, Fk**
Rj,Rk:	**Flags indicating when Fj, Fk are ready**

- **Register result status—Indicates which functional unit will write each register, if one exists. Blank when no pending instructions will write that register**

Scoreboard Example

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2				
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

Functional unit status:

Time Name	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
Integer	No								
Mult1	No								
Mult2	No								
Add	No								
Divide	No								

Register result status:

Clock	*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
FU									

Detailed Scoreboard Pipeline Control

Instruction status	Wait until	Bookkeeping
Issue	Not busy (FU) and not Result(D)	Busy(FU)← yes; Op(FU)← op; Fi(FU)← `D'; Fj(FU)← `S1'; Fk(FU)← `S2'; Qj← Result('S1'); Qk← Result(`S2'); Rj← not Qj; Rk← not Qk; Result('D')← FU;
Read operands	Rj and Rk	Rj← No; Rk← No; Qj ← 0; Qk ← 0
Execution complete	Functional unit done	
Write result	∀f((Fj(f)≠Fi(FU) or Rj(f)=No) & (Fk(f)≠Fi(FU) or Rk(f)=No))	∀f(if Qj(f)=FU then Rj(f)← Yes); ∀f(if Qk(f)=FU then Rk(f)← Yes); Result(Fi(FU))← 0; Busy(FU)← No

Scoreboard Example: Cycle 1

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1			
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	Yes	Load	F6		R2			Yes	Yes
	Mult1	No								
	Mult2	No								
	Add	No								
	Divide	No								

Register result status:

Clock **1**

	F0	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
FU				Integer					

Scoreboard Example: Cycle 2

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2		
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	Yes	Load	F6		R2			No	No
	Mult1	No								
	Mult2	No								
	Add	No								
	Divide	No								

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
2	*FU*				Integer					

- **Issue 2nd LD?**

Scoreboard Example: Cycle 3

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	Yes	Load	F6		R2			No	No
	Mult1	No								
	Mult2	No								
	Add	No								
	Divide	No								

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
3	*FU*				Integer					

- **Issue MULT?**

Scoreboard Example: Cycle 4

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
	Mult1	No								
	Mult2	No								
	Add	No								
	Divide	No								

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
4	*FU*				0					

Scoreboard Example: Cycle 5

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5			
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

Functional unit status:

Time Name	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
Integer	Yes	Load	F2		R3			Yes	Yes
Mult1	No								
Mult2	No								
Add	No								
Divide	No								

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
5	*FU*		Integer							

Scoreboard Example: Cycle 6

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6		
MULTD	F0	F2	F4	6			
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	Yes	Load	F2		R3			No	No
	Mult1	Yes	Mult	F0	F2	F4	Integer	0	No	Yes
	Mult2	No								
	Add	No								
	Divide	No								

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
6	*FU*	Mult1	Integer							

Scoreboard Example: Cycle 7

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	
MULTD	F0	F2	F4	6			
SUBD	F8	F6	F2	7			
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	Yes	Load	F2		R3			No	No
	Mult1	Yes	Mult	F0	F2	F4	Integer		No	Yes
	Mult2	No								
	Add	Yes	Sub	F8	F6	F2		Integer	Yes	No
	Divide	No								

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
7	*FU*	Mult1	Integer			Add				

- **Read multiply operands?**

Scoreboard Example: Cycle 8a
(First half of clock cycle)

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	
MULTD	F0	F2	F4	6			
SUBD	F8	F6	F2	7			
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	Yes	Load	F2		R3			No	No
	Mult1	Yes	Mult	F0	F2	F4	Integer		No	Yes
	Mult2	No								
	Add	Yes	Sub	F8	F6	F2		Integer	Yes	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
8	*FU*	Mult1	Integer			Add	Divide			

Scoreboard Example: Cycle 8b
(Second half of clock cycle)

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6			
SUBD	F8	F6	F2	7			
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
	Mult1	Yes	Mult	F0	F2	F4			Yes	Yes
	Mult2	No								
	Add	Yes	Sub	F8	F6	F2			Yes	Yes
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
8	*FU*	Mult1				Add	Divide			

Scoreboard Example: Cycle 9

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9		
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2				

Functional unit status:

Note Remaining

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
10	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
2	Add	Yes	Sub	F8	F6	F2			No	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
9	*FU*	Mult1				Add	Divide			

- **Read operands for MULT & SUB?**
- **Issue ADDD?**

Scoreboard Example: Cycle 10

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9		
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
9	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
1	Add	Yes	Sub	F8	F6	F2			No	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
10	*FU*	Mult1				Add	Divide			

Scoreboard Example: Cycle 11

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
8	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
0	Add	Yes	Sub	F8	F6	F2			No	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
11	*FU*	Mult1				Add	Divide			

Scoreboard Example: Cycle 12

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2				

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
7	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
	Add	No								
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
12	*FU*	Mult1					Divide			

- **Read operands for DIVD?**

Scoreboard Example: Cycle 13

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2	13			

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
6	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
	Add	Yes	Add	F6	F8	F2			Yes	Yes
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
13	*FU*	Mult1			Add		Divide			

Scoreboard Example: Cycle 14

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2	13	14		

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
5	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
2	Add	Yes	Add	F6	F8	F2			No	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
14	*FU*	Mult1			Add		Divide			

Scoreboard Example: Cycle 15

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2	13	14		

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
4	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
1	Add	Yes	Add	F6	F8	F2			No	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
15	*FU*	Mult1			Add		Divide			

Scoreboard Example: Cycle 16

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2	13	14	16	

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
3	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
0	Add	Yes	Add	F6	F8	F2			No	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
16	*FU*	Mult1			Add		Divide			

Scoreboard Example: Cycle 17

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2	13	14	16	

WAR Hazard!

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
2	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
	Add	Yes	Add	F6	F8	F2			No	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
17	*FU*	Mult1			Add		Divide			

- **Why not write result of ADDD???**

Scoreboard Example: Cycle 18

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2	13	14	16	

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
1	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
	Add	Yes	Add	F6	F8	F2			No	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
18	*FU*	Mult1			Add		Divide			

Scoreboard Example: Cycle 19

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2	13	14	16	

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
0	Mult1	Yes	Mult	F0	F2	F4			No	No
	Mult2	No								
	Add	Yes	Add	F6	F8	F2			No	No
	Divide	Yes	Div	F10	F0	F6	Mult1		No	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
19	*FU*	Mult1			Add		Divide			

Scoreboard Example: Cycle 20

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2	13	14	16	

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
	Mult1	No								
	Mult2	No								
	Add	Yes	Add	F6	F8	F2			No	No
	Divide	Yes	Div	F10	F0	F6			Yes	Yes

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
20	*FU*				Add		Divide			

Scoreboard Example: Cycle 21

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8	21		
ADDD	F6	F8	F2	13	14	16	

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
	Mult1	No								
	Mult2	No								
	Add	Yes	Add	F6	F8	F2			No	No
	Divide	Yes	Div	F10	F0	F6			No	No

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
21	*FU*				Add		Divide			

- **WAR Hazard is now gone...**

Scoreboard Example: Cycle 22

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8	21		
ADDD	F6	F8	F2	13	14	16	22

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
	Mult1	No								
	Mult2	No								
	Add	No								
39	Divide	Yes	Div	F10	F0	F6			No	No

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
22	*FU*						Divide			

Faster than light computation (skip a couple of cycles)

Scoreboard Example: Cycle 61

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8	21	61	
ADDD	F6	F8	F2	13	14	16	22

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
	Mult1	No								
	Mult2	No								
	Add	No								
0	Divide	Yes	Div	F10	F0	F6			No	No

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
61	*FU*						Divide			

Scoreboard Example: Cycle 62

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8	21	61	62
ADDD	F6	F8	F2	13	14	16	22

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
	Mult1	No								
	Mult2	No								
	Add	No								
	Divide	No								

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
62	*FU*									

Review: Scoreboard Example: Cycle 62

Instruction status:

Instruction		*j*	*k*	*Issue*	*Read Oper*	*Exec Comp*	*Write Result*
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8	21	61	62
ADDD	F6	F8	F2	13	14	16	22

Functional unit status:

Time	*Name*	*Busy*	*Op*	*dest Fi*	*S1 Fj*	*S2 Fk*	*FU Qj*	*FU Qk*	*Fj? Rj*	*Fk? Rk*
	Integer	No								
	Mult1	No								
	Mult2	No								
	Add	No								
	Divide	No								

Register result status:

Clock		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
62	*FU*									

- **In-order issue; out-of-order execute & commit**

Limitations of CDC 6600 Scoreboard

- **No forwarding hardware**
- **Limited to the number of scoreboard entries**
- **Limited to the number of functional units (structural hazards), especially integer/load store units**
- **Do not issue on structural hazards**
- **Wait for WAR hazards**
- **Prevent WAW hazards**

Summary

- **Increasingly powerful (and complex) dynamic mechanism for detecting and resolving hazards**
  - **In-order pipeline, in-order op-fetch with register reservations, in-order issue with scoreboard**
  - **Allow later instructions to proceed around ones that are stalled**
  - **Facilitate multiple issue**

- **Compiler techniques make it easier for HW to find the ILP**
  - **Reduces the impact of more sophisticated organization**
  - **Requires a larger architected namespace**
  - **Easier for more structured code**